

数学分析(3) 期末试题解答参考 (2023-01-11)

一(15分) 对于二维闭方块 $[0, 1]^2$ 及其边界 $\partial[0, 1]^2$, 叙述Green公式并直接证明之.

【解】所指的Green 公式为: 若数值函数 $P, Q \in C^1([0, 1]^2)$, 则沿着 $\partial[0, 1]^2$ 的逆时针方向积分有

$$\int_{\partial[0, 1]^2} Pdx + Qdy = \int_{[0, 1]^2} \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy.$$

[证]: 沿着 $\partial[0, 1]^2$ 的逆时针方向积分, 由积分的可加性有

$$\begin{aligned} \int_{\partial[0, 1]^2} Pdx + Qdy &= \int_{[0, 1] \times \{0\}} + \int_{\{1\} \times [0, 1]} + \int_{[0, 1] \times \{1\}} + \int_{\{0\} \times [0, 1]} \\ &= \int_{[0, 1] \times \{0\}} Pdx + \int_{\{1\} \times [0, 1]} Qdy + \int_{[0, 1] \times \{1\}} Pdx + \int_{\{0\} \times [0, 1]} Qdy \\ &= \int_0^1 P(x, 0)dx + \int_0^1 Q(1, y)dy - \int_0^1 P(x, 1)dx - \int_0^1 Q(0, y)dy \\ &= \int_0^1 (Q(1, y) - Q(0, y))dy - \int_0^1 (P(x, 1) - P(x, 0))dx \\ &= \int_0^1 \left(\int_0^1 \frac{\partial Q}{\partial x}(x, y)dx \right) dy - \int_0^1 \left(\int_0^1 \frac{\partial P}{\partial y}(x, y)dy \right) dx \\ &= \int_0^1 \int_0^1 \frac{\partial Q}{\partial x}(x, y)dx dy - \int_0^1 \int_0^1 \frac{\partial P}{\partial y}(x, y)dy dx \\ &= \int_{[0, 1] \times [0, 1]} \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy. \end{aligned}$$

以上用到重积分的累次积分换序定理(Fubini定理). □

二(15分) 设 $D \subset \mathbb{R}^2$ 为一个有界单连通开集, 其边界 ∂D 分片光滑. 设实值函数 $f, g \in C^2(\overline{D})$ 满足Cauchy-Riemann 方程

$$f'_x(x, y) = g'_y(x, y), \quad f'_y(x, y) = -g'_x(x, y) \quad \forall (x, y) \in \overline{D}.$$

其中 $f'_x = \frac{\partial f}{\partial x}$, $f'_y = \frac{\partial f}{\partial y}$. 类似地简记 $f''_{xy} = (f'_x)'_y = \frac{\partial^2 f}{\partial x \partial y}$, 等等. 证明: 沿着 $\partial \overline{D}$ 的逆时针方向积分有

$$\int_{\partial \overline{D}} f(x, y)dg(x, y) - g(x, y)df(x, y) \geq 0.$$

然后证明: 若等号“=0”成立, 则 f, g 在 $\overline{D}$ 上为常值函数.

【证】计算

$$\begin{aligned} f dg - g df &= f(g'_x dx + g'_y dy) - g(f'_x dx + f'_y dy) \\ &= f g'_x dx + f g'_y dy - g f'_x dx - g f'_y dy = (f g'_x - g f'_x) dx + (f g'_y - g f'_y) dy. \end{aligned}$$

应用Green 公式有

$$\begin{aligned} \int_{\partial\bar{D}} f(x,y)dg(x,y) - g(x,y)df(x,y) &= \int_{\partial\bar{D}} (fg'_x - gf'_x)dx + (fg'_y - gf'_y)dy \\ &= \int_{\bar{D}} \left(\frac{fg'_y - gf'_y}{\partial x} - \frac{fg'_x - gf'_x}{\partial y} \right) dx dy = \int_{\bar{D}} \left((fg'_y - gf'_y)'_x - (fg'_x - gf'_x)'_y \right) dx dy \\ &= \int_{\bar{D}} (f'_x g'_y + f g''_{yx} - g'_x f'_y - g f''_{yx} - f'_y g'_x - f g''_{xy} + g'_y f'_x + g f''_{xy}) dx dy \\ &= 2 \int_{\bar{D}} (f'_x g'_y - g'_x f'_y) dx dy \end{aligned}$$

其中用到 C^2 -类函数的混合偏导数可交换: $f''_{xy} = f''_{yx}$.

或者: 由 $f, g \in C^2(\bar{D})$ 和外微分运算有

$$\begin{aligned} d(fdg - gdf) &= (df) \wedge (dg) + f d^2g - (dg) \wedge (df) - g d^2f \\ &= 2(df) \wedge (dg) = 2 \frac{\partial(f, g)}{\partial(x, y)} dx \wedge dy. \end{aligned}$$

因此据Stokes 公式有

$$\int_{\partial\bar{D}} fdg - gdf = \int_{\bar{D}} d(fdg - gdf) = 2 \int_{\bar{D}} \frac{\partial(f, g)}{\partial(x, y)} dx \wedge dy = 2 \int_{\bar{D}} \frac{\partial(f, g)}{\partial(x, y)} dx dy.$$

因 f, g 满足Cauchy-Riemann 方程, 故

$$\frac{\partial(f, g)}{\partial(x, y)} = f'_x g'_y - g'_x f'_y = (f'_x)^2 + (f'_y)^2 = |\nabla f|^2.$$

所以

$$\int_{\partial\bar{D}} f(x,y)dg(x,y) - g(x,y)df(x,y) = 2 \int_{\bar{D}} |\nabla f|^2 dx dy \geq 0.$$

最后假设

$$\int_{\partial\bar{D}} f(x,y)dg(x,y) - g(x,y)df(x,y) = 0 \quad \text{即} \quad \int_{\bar{D}} |\nabla f|^2 dx dy = 0.$$

则由 $|\nabla f|^2 \geq 0$ 且连续知 $|\nabla f|^2 \equiv 0$ 于 $\bar{D}$, 即 $\nabla f \equiv \mathbf{0}$ 于 $\bar{D}$. 因 $\bar{D}$ 连通, 故由数学分析(2)知 f 在 $\bar{D}$ 上为常值函数. 又由 $|\nabla g|^2 \equiv |\nabla f|^2$ 知 g 在 $\bar{D}$ 上也为常值函数. $\square$

三(15分). 计算流速为 $\mathbf{F} = \mathbf{F}(x, y, z) = (x^2 - y^2, y^2 - z^2, z^2 - x^2)$ 的流体通过上半球面 $S = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 = 1, z \geq 0\}$ 的流量

$$\int_S \langle \mathbf{F}, \mathbf{n} \rangle d\sigma = ?$$

其中 $\mathbf{n} = \mathbf{n}(x, y, z)$ 是 S 上的单位法向量场, 其指向与 z -轴的正向成锐角. (建议: 解题时画出曲面和定向的草图.)

【解】作平面片 $S_0 = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 \leq 1, z = 0\}$. 则 $S \cup S_0 = \partial\bar{\Omega}$ 其中 $\bar{\Omega} = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 \leq 1, z \geq 0\}$. 由Gauss 散度定理, 沿着 $\partial\bar{\Omega}$ 朝外方向的积分有

$$\int_S \langle \mathbf{F}, \mathbf{n} \rangle d\sigma + \int_{S_0} \langle \mathbf{F}, \mathbf{n} \rangle d\sigma = \int_{S \cup S_0} \langle \mathbf{F}, \mathbf{n} \rangle d\sigma = \int_{\partial\bar{\Omega}} \langle \mathbf{F}, \mathbf{n} \rangle d\sigma$$

$$= \int_{\Omega} \nabla \cdot \mathbf{F} dx dy dz = \int_{x^2+y^2+z^2, z \leq 0} (2x + 2y + 2z) dx dy dz$$

$\Rightarrow$

$$\int_S \langle \mathbf{F}, \mathbf{n} \rangle d\sigma = - \int_{S_0} \langle \mathbf{F}, \mathbf{n} \rangle d\sigma + \int_{x^2+y^2+z^2, z \leq 0} (2x + 2y + 2z) dx dy dz.$$

计算: 对应于 $\partial\Omega$ 朝外的方向, S_0 的方向为朝下. 因此 S_0 上的单位法向量为 $\mathbf{n} = (0, 0, -1)$. 于是有

$$\begin{aligned} - \int_{S_0} \langle \mathbf{F}, \mathbf{n} \rangle d\sigma &= - \int_{S_0} (z^2 - x^2)(-1) d\sigma = - \int_{S_0} x^2 d\sigma = - \int_{x^2+y^2 \leq 1} x^2 dx dy \\ &= - \frac{1}{2} \int_{x^2+y^2 \leq 1} (x^2 + y^2) dx dy = - \frac{1}{2} \int_0^1 r^3 dr \cdot 2\pi = \left(-\frac{1}{2}\right) \cdot \frac{1}{4} \cdot 2\pi = -\frac{\pi}{4}. \end{aligned}$$

又由被积函数的奇偶性和坐标换元有

$$\begin{aligned} \int_{x^2+y^2+z^2 \leq 1, z \geq 0} (2x + 2y + 2z) dx dy dz &= 2 \int_{x^2+y^2+z^2 \leq 1, z \geq 0} z dx dy dz \\ &= 2 \int_0^1 z \left(\iint_{x^2+y^2 \leq 1-z^2} dx dy \right) dz \\ &= 2 \int_0^1 z\pi(1-z^2) dz = 2\pi \int_0^1 (z - z^3) dz = 2\pi \left(\frac{1}{2} - \frac{1}{4} \right) = \frac{\pi}{2}. \end{aligned}$$

所以

$$\int_S \langle \mathbf{F}, \mathbf{n} \rangle d\sigma = -\frac{\pi}{4} + \frac{\pi}{2} = \frac{\pi}{4}.$$

□

四(10分). 设空间 $\mathbb{R}^3$ 中有一个 C^1 -类的无旋的力场 $\mathbf{F} = (P(x, y, z), Q(x, y, z), R(x, y, z))$ (无旋指 $\text{rot}\mathbf{F} = \mathbf{0}$ 于 $\mathbb{R}^3$), 其竖直分力 $R(x, y, z) = e^{x+y+z}, (x, y, z) \in \mathbb{R}^3$. 设有一个单位质量的质点沿着曲线 $\gamma = \{(t(1-t), t^2(1-t), t^3) \mid t \in [0, 1]\}$ 从原点 $(0, 0, 0)$ 移动到点 $(0, 0, 1)$. 试计算在这个过程中力场 $\mathbf{F} = (P, Q, R)$ 对该质点所做的功

$$W = \int_{\gamma} P dx + Q dy + R dz = ?$$

【解】 因 $\mathbf{F}$ 无旋且 $\mathbb{R}^3$ 为单连通区域, 故由场论基本定理知上述做功(即道路积分)只与路径的起点和终点有关而与路径的中途无关. 因此上述积分等于被积函数在直线段 $\{(0, 0)\} \times [0, 1]$ 上的积分, 所以

$$W = \int_{\{(0,0)\} \times [0,1]} P dx + Q dy + R dz = \int_0^1 R(0, 0, z) dz = \int_0^1 e^z dz = e - 1.$$

□

五(15分) 设 $\mathcal{H}$ 为一个Hilbert空间, $\{e_k\}_{k=1}^{\infty}$ 是 $\mathcal{H}$ 中的一个ON系. 设 $\langle f, g \rangle, \|f\| = \sqrt{\langle f, f \rangle}$ 为 $\mathcal{H}$ 上的内积和范数. 令

$$S_n(f) = \sum_{k=1}^n \langle f, e_k \rangle e_k, \quad f \in \mathcal{H}, \quad n = 1, 2, 3, \dots$$

证明:

$$\{e_k\}_{k=1}^{\infty} \text{ 是 } \mathcal{H} \text{ 的 ON 基 } \iff \lim_{n \rightarrow \infty} \left\| f - \frac{1}{n} \sum_{k=1}^n S_k(f) \right\| = 0 \quad \forall f \in \mathcal{H}.$$

【证】“ $\implies$ ”：设 $\{e_n\}_{n=1}^{\infty}$ 是 $\mathcal{H}$ 的 ON 基. 则由 ON 基的定义知

$$f = \sum_{k=1}^{\infty} \langle f, e_k \rangle e_k = \lim_{n \rightarrow \infty} \sum_{k=1}^n \langle f, e_k \rangle e_k \quad \text{in norm} \quad \forall f \in \mathcal{H}$$

即

$$\lim_{n \rightarrow \infty} \|f - S_n(f)\| = 0 \quad \forall f \in \mathcal{H}.$$

由此便有

$$\left\| f - \frac{1}{n} \sum_{k=1}^n S_k(f) \right\| = \left\| \frac{1}{n} \sum_{k=1}^n (f - S_k(f)) \right\| \leq \frac{1}{n} \sum_{k=1}^n \|f - S_k(f)\| \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

“ $\impliedby$ ”：由假设知 $\{e_k\}_{k=1}^{\infty}$ 的有限线性组合 $\sum_{k=1}^n c_k e_k$ 的全体在 $\mathcal{H}$ 中稠密(按 $\mathcal{H}$ 的范数). 因此据 Hilbert 空间的 ON 基的刻画定理知 $\{e_k\}_{k=1}^{\infty}$ 是 $\mathcal{H}$ 的 ON 基.

或者: 对任意 $f \in \mathcal{H}$, 由广义 Fourier 级数的部分和 $S_n(f)$ 的最佳逼近性质和假设条件有

$$\|f - S_n(f)\| \leq \left\| f - \frac{1}{n} \sum_{k=1}^n S_k(f) \right\| \rightarrow 0 \quad (n \rightarrow \infty).$$

所以由 ON 基的定义知 $\{e_k\}_{k=1}^{\infty}$ 是 $\mathcal{H}$ 的 ON 基. 最后这个证明不需要 $\mathcal{H}$ 的完备性.

或者: 对任意 $f \in \mathcal{H}$, 假设 $\langle f, e_k \rangle = 0 \quad \forall k = 1, 2, 3, \dots$, 则有

$$\sum_{k=1}^n S_k(f) = 0, \quad k = 1, 2, 3, \dots$$

从而由假设有

$$\|f\| = \left\| f - \frac{1}{n} \sum_{k=1}^n S_k(f) \right\| \rightarrow 0 \quad (n \rightarrow \infty).$$

因此 $\|f\| = 0$ 即 $f = 0$. 据 Hilbert 空间的 ON 基的刻画定理知 $\{e_k\}_{k=1}^{\infty}$ 是 $\mathcal{H}$ 的 ON 基. □

六(15分). 设实值或复值函数 $f \in C^1([-\pi, \pi])$ 满足 $f(\pi) = f(-\pi)$. 证明

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} |f'(x)|^2 dx \geq \frac{1}{2\pi} \int_{-\pi}^{\pi} |f(x)|^2 dx - \left| \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx \right|^2.$$

然后证明若等号成立, 则 $f(x) = C + A \cos(x) + B \sin(x)$, $x \in [-\pi, \pi]$, 其中 A, B, C 为常数.

【证】首先由假设易见 $f, f' \in L^2([-\pi, \pi])$, 因此由 Parseval 等式有

$$\begin{aligned} \frac{1}{2\pi} \int_{-\pi}^{\pi} |f(x)|^2 dx &= \sum_{k \in \mathbb{Z}} |\widehat{f}(k)|^2, \\ \frac{1}{2\pi} \int_{-\pi}^{\pi} |f'(x)|^2 dx &= \sum_{k \in \mathbb{Z}} |\widehat{f'}(k)|^2 \end{aligned}$$

其中

$$\widehat{f}(k) = \langle f, \frac{e^{ik\cdot}}{\sqrt{2\pi}} \rangle \frac{1}{\sqrt{2\pi}} = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-ikx} dx, \quad k \in \mathbb{Z}.$$

计算: 由分部积分和 $f(\pi) = f(-\pi)$ 有

$$\begin{aligned}\widehat{f}'(k) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f'(x) e^{-ikx} dx = \frac{1}{2\pi} \left(f(x) e^{-ikx} \Big|_{x=-\pi}^{x=\pi} - \int_{-\pi}^{\pi} f(x) e^{-ikx} (-ik) dx \right) \\ &= ik \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-ikx} dx = ik \widehat{f}(k), \quad k \in \mathbb{Z}.\end{aligned}$$

所以

$$\begin{aligned}\frac{1}{2\pi} \int_{-\pi}^{\pi} |f'(x)|^2 dx &= \sum_{k \in \mathbb{Z}} |\widehat{f}'(k)|^2 = \sum_{k \in \mathbb{Z}} |\widehat{f}(k)|^2 |k|^2 = \sum_{0 \neq k \in \mathbb{Z}} |\widehat{f}(k)|^2 |k|^2 \\ &\implies \\ \frac{1}{2\pi} \int_{-\pi}^{\pi} |f'(x)|^2 dx &- \left(\frac{1}{2\pi} \int_{-\pi}^{\pi} |f(x)|^2 dx - \left| \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx \right|^2 \right) \\ &= \sum_{0 \neq k \in \mathbb{Z}} |\widehat{f}(k)|^2 |k|^2 - \left(\sum_{k \in \mathbb{Z}} |\widehat{f}(k)|^2 - |\widehat{f}(0)|^2 \right) \\ &= \sum_{0 \neq k \in \mathbb{Z}} |\widehat{f}(k)|^2 |k|^2 - \sum_{0 \neq k \in \mathbb{Z}} |\widehat{f}(k)|^2 = \sum_{0 \neq k \in \mathbb{Z}} |\widehat{f}(k)|^2 (|k|^2 - 1) \geq 0.\end{aligned}$$

若等号 “= 0” 成立, 则 $|\widehat{f}(k)| = 0$ for all $|k| \geq 2$. 这蕴含

$$\sum_{|k| \leq n} \widehat{f}(k) e^{ikx} = \sum_{|k| \leq 1} \widehat{f}(k) e^{ikx} \quad \forall n \geq 2.$$

于是由 L^2 -收敛有

$$0 = \lim_{n \rightarrow \infty} \int_{-\pi}^{\pi} |f(x) - \sum_{|k| \leq n} \widehat{f}(k) e^{ikx}|^2 dx = \int_{-\pi}^{\pi} |f(x) - \sum_{|k| \leq 1} \widehat{f}(k) e^{ikx}|^2 dx.$$

所以

$$f(x) \equiv \sum_{|k| \leq 1} \widehat{f}(k) e^{ikx} = \widehat{f}(0) + \widehat{f}(1) e^{ix} + \widehat{f}(-1) e^{-ix}, \quad x \in [-\pi, \pi].$$

这等价于 $f(x) \equiv C + A \cos(x) + B \sin(x)$, $x \in [-\pi, \pi]$, 其中 A, B, C 为常数. □

七(15分). 设 $n \geq 2$. 考虑一个2次微分形式

$$\omega(x) = \sum_{1 \leq i < j \leq n} a_{ij}(x) dx^i \wedge dx^j, \quad x = (x^1, x^2, \dots, x^n) \in \mathbb{R}^n.$$

其中系数函数 $a_{ij} \in C^1(\mathbb{R}^n)$, $1 \leq i < j \leq n$. 假设 ω 有一个 C^2 -类的原函数, 即存在 $F^i \in C^2(\mathbb{R}^n)$, $i = 1, 2, \dots, n$, 使得1次微分形式 $\eta(x) = \sum_{i=1}^n F^i(x) dx^i$ 满足 $d\eta = \omega$ (意即 ω 是一个 C^1 -类的恰当形式). 证明: 存在函数 $b_j^i \in C^1(\mathbb{R}^n)$, $i, j = 1, 2, \dots, n$, 使得

$$a_{ij}(x) = b_j^i(x) - b_i^j(x) \quad \forall x \in \mathbb{R}^n, \quad \forall 1 \leq i < j \leq n \quad (1)$$

$$\text{且} \quad \frac{\partial b_k^i(x)}{\partial x^j} = \frac{\partial b_j^i(x)}{\partial x^k} \quad \forall x \in \mathbb{R}^n, \quad \forall i, j, k \in \{1, 2, \dots, n\}. \quad (2)$$

【证】如上, 设 $\eta(x) = \sum_{i=1}^n F^i(x)dx^i$, $F^i \in C^2(\mathbb{R}^n)$, $i = 1, 2, \dots, n$. 则由外微分运算法则有

$$\begin{aligned} d\eta(x) &= d\left(\sum_{i=1}^n F^i(x)dx^i\right) = \sum_{i=1}^n dF^i(x) \wedge dx^i \\ &= \sum_{i=1}^n \sum_{j=1}^n \frac{\partial F^i(x)}{\partial x^j} dx^j \wedge dx^i = \sum_{1 \leq i < j \leq n} \frac{\partial F^i(x)}{\partial x^j} dx^j \wedge dx^i + \sum_{1 \leq j < i \leq n} \frac{\partial F^i(x)}{\partial x^j} dx^j \wedge dx^i \\ &= - \sum_{1 \leq i < j \leq n} \frac{\partial F^i(x)}{\partial x^j} dx^i \wedge dx^j + \sum_{1 \leq i < j \leq n} \frac{\partial F^j(x)}{\partial x^i} dx^i \wedge dx^j \\ &= \sum_{1 \leq i < j \leq n} \left(\frac{\partial F^j(x)}{\partial x^i} - \frac{\partial F^i(x)}{\partial x^j} \right) dx^i \wedge dx^j \quad \forall x \in \mathbb{R}^n. \end{aligned}$$

因此结合假设 $\omega = d\eta$ 得到

$$\sum_{1 \leq i < j \leq n} a_{ij}(x) dx^i \wedge dx^j = \sum_{1 \leq i < j \leq n} \left(\frac{\partial F^j(x)}{\partial x^i} - \frac{\partial F^i(x)}{\partial x^j} \right) dx^i \wedge dx^j \quad \forall x \in \mathbb{R}^n.$$

据微分形式最简表示的唯一性, 这蕴含

$$a_{ij}(x) = \frac{\partial F^j(x)}{\partial x^i} - \frac{\partial F^i(x)}{\partial x^j} \quad \forall x \in \mathbb{R}^n, \quad \forall 1 \leq i < j \leq n.$$

取

$$b_j^i(x) = \frac{\partial F^i(x)}{\partial x^j}, \quad x \in \mathbb{R}^n, \quad i, j \in \{1, 2, \dots, n\}.$$

则有 $b_j^i \in C^1(\mathbb{R}^n)$, $i, j = 1, 2, \dots, n$

$$a_{ij}(x) = b_j^i(x) - b_i^j(x) \quad \forall x \in \mathbb{R}^n, \quad \forall 1 \leq i < j \leq n$$

并由 $F^i \in C^2(\mathbb{R}^n)$ 从而偏导数可换序有

$$\frac{\partial b_k^i(x)}{\partial x^j} = \frac{\partial}{\partial x^j} \left(\frac{\partial F^i(x)}{\partial x^k} \right) = \frac{\partial}{\partial x^k} \left(\frac{\partial F^i(x)}{\partial x^j} \right) = \frac{\partial b_j^i(x)}{\partial x^k} \quad \forall x \in \mathbb{R}^n, \quad \forall i, j, k \in \{1, 2, \dots, n\}.$$

□

八(附加题10分). 证明第七题中的两个必要条件(1),(2) (其中 $b_j^i \in C^1(\mathbb{R}^n)$, $i, j = 1, 2, \dots, n$) 也是使得 ω 有一个 C^2 -类的原函数 $\eta(x) = \sum_{i=1}^n F^i(x)dx^i$ 的充分条件.

【证】设存在函数组 $b_j^i \in C^1(\mathbb{R}^n)$, $i, j = 1, 2, \dots, n$ 满足条件(1),(2).

[证法1.] 考虑1次形式

$$\omega^i(x) = \sum_{j=1}^n b_j^i(x) dx^j, \quad x \in \mathbb{R}^n; \quad i = 1, 2, \dots, n.$$

由外微分运算法则和条件(2) 有

$$d\omega^i(x) = \sum_{j=1}^n d(b_j^i(x) dx^j) = \sum_{j=1}^n (db_j^i(x)) \wedge dx^j$$

$$\begin{aligned}
&= \sum_{j=1}^n \sum_{k=1}^n \frac{\partial b_j^i(x)}{\partial x^k} dx^k \wedge dx^j = \sum_{1 \leq j < k \leq n} \frac{\partial b_j^i(x)}{\partial x^k} dx^k \wedge dx^j + \sum_{1 \leq k < j \leq n} \frac{\partial b_j^i(x)}{\partial x^k} dx^k \wedge dx^j \\
&= \sum_{1 \leq j < k \leq n} \frac{\partial b_j^i(x)}{\partial x^k} dx^k \wedge dx^j + \sum_{1 \leq j < k \leq n} \frac{\partial b_k^i(x)}{\partial x^j} dx^j \wedge dx^k \\
&= \sum_{1 \leq j < k \leq n} \left(\frac{\partial b_k^i(x)}{\partial x^j} - \frac{\partial b_j^i(x)}{\partial x^k} \right) dx^j \wedge dx^k = 0, \quad x \in \mathbb{R}^n.
\end{aligned}$$

所以 ω^i 在 $\mathbb{R}^n$ 上是闭形式. 因 $\mathbb{R}^n$ 为星形区域, 故由庞加莱引理知 ω^i 还是恰当形式, 因此存在 $F^i \in C^2(\mathbb{R}^n)$ 使得 $dF^i(x) = \omega^i(x), i = 1, 2, \dots, n$. 这里 $F^i \in C^2(\mathbb{R}^n)$ 是因为 ω^i 是 C^1 -类的微分形式. 以上得到

$$\sum_{j=1}^n \frac{\partial F^i(x)}{\partial x^j} dx^j = dF^i(x) = \omega^i(x) = \sum_{j=1}^n b_j^i(x) dx^j, \quad x \in \mathbb{R}^n; \quad i = 1, 2, \dots, n.$$

由微分形式的唯一性知

$$\frac{\partial F^i(x)}{\partial x^j} = b_j^i(x), \quad x \in \mathbb{R}^n; \quad i, j = 1, 2, \dots, n.$$

据此和条件(1)便知对于 C^2 -类的1次形式 $\eta(x) = \sum_{i=1}^n F^i(x) dx^i$ 有

$$\begin{aligned}
d\eta(x) &= \sum_{1 \leq i < j \leq n} \left(\frac{\partial F^j(x)}{\partial x^i} - \frac{\partial F^i(x)}{\partial x^j} \right) dx^i \wedge dx^j \\
&= \sum_{1 \leq i < j \leq n} \left(b_i^j(x) - b_j^i(x) \right) dx^i \wedge dx^j = \sum_{1 \leq i < j \leq n} a_{ij}(x) dx^i \wedge dx^j \\
&= \omega(x) \quad \forall x \in \mathbb{R}^n.
\end{aligned}$$

所以 η 是 ω 的一个原函数.

[证法2.] 若能证明 ω 是 $\mathbb{R}^n$ 上的闭形式, 则由 $\mathbb{R}^n$ 是星形区域和庞加莱引理知 ω 是 $\mathbb{R}^n$ 上的恰当形式, 即存在 $\mathbb{R}^n$ 上的一个微分形式 η 使得 $d\eta = \omega$. 因 ω 是 C^1 类的2次微分形式, 故 η 是 C^2 -类的1次微分形式: $\eta(x) = \sum_{i=1}^n F^i(x) dx^i$ 其中 $F^i \in C^2(\mathbb{R}^n), i = 1, 2, \dots, n$.

所以下面只需证明 ω 是闭形式. 计算 $d\omega$: 令

$$A_{i,j,k}(x) = \frac{\partial b_j^i(x)}{\partial x^k}, \quad x \in \mathbb{R}^n, \quad i, j, k \in \{1, 2, \dots, n\}.$$

则由条件(1)有

$$\begin{aligned}
d\omega &= \sum_{1 \leq i < j \leq n} \sum_{k=1}^n \left(\frac{\partial b_i^j(x)}{\partial x^k} - \frac{\partial b_j^i(x)}{\partial x^k} \right) dx^k \wedge dx^i \wedge dx^j \\
&= \sum_{1 \leq i < j \leq n} \sum_{k=1}^n A_{j,i,k} dx^k \wedge dx^i \wedge dx^j - \sum_{1 \leq i < j \leq n} \sum_{k=1}^n A_{i,j,k} dx^k \wedge dx^i \wedge dx^j \\
&= I_1 - I_2;
\end{aligned}$$

I_1

$$\begin{aligned}
&= \sum_{1 \leq i < j < k \leq n} A_{j,i,k} dx^k \wedge dx^i \wedge dx^j + \sum_{1 \leq i < k < j \leq n} A_{j,i,k} dx^k \wedge dx^i \wedge dx^j + \sum_{1 \leq k < i < j \leq n} A_{j,i,k} dx^k \wedge dx^i \wedge dx^j \\
&= \sum_{1 \leq i < j < k \leq n} A_{j,i,k} dx^i \wedge dx^j \wedge dx^k - \sum_{1 \leq i < k < j \leq n} A_{j,i,k} dx^i \wedge dx^k \wedge dx^j + \sum_{1 \leq k < i < j \leq n} A_{j,i,k} dx^k \wedge dx^i \wedge dx^j \\
&= \sum_{1 \leq i < j < k \leq n} A_{j,i,k} dx^i \wedge dx^j \wedge dx^k - \sum_{1 \leq i < j < k \leq n} A_{k,i,j} dx^i \wedge dx^k \wedge dx^j + \sum_{1 \leq i < j < k \leq n} A_{k,j,i} dx^i \wedge dx^j \wedge dx^k \\
&= \sum_{1 \leq i < j < k \leq n} (A_{j,i,k} - A_{k,i,j} + A_{k,j,i}) dx^i \wedge dx^j \wedge dx^k,
\end{aligned}$$

$$\begin{aligned}
I_2 &= \sum_{1 \leq i < j < k \leq n} A_{i,j,k} dx^k \wedge dx^i \wedge dx^j + \sum_{1 \leq i < k < j \leq n} A_{i,j,k} dx^k \wedge dx^i \wedge dx^j + \sum_{1 \leq k < i < j \leq n} A_{i,j,k} dx^k \wedge dx^i \wedge dx^j \\
&= \sum_{1 \leq i < j < k \leq n} A_{i,j,k} dx^i \wedge dx^j \wedge dx^k - \sum_{1 \leq i < k < j \leq n} A_{i,j,k} dx^i \wedge dx^k \wedge dx^j + \sum_{1 \leq k < i < j \leq n} A_{i,j,k} dx^k \wedge dx^i \wedge dx^j \\
&= \sum_{1 \leq i < j < k \leq n} A_{i,j,k} dx^i \wedge dx^j \wedge dx^k - \sum_{1 \leq i < j < k \leq n} A_{i,k,j} dx^i \wedge dx^j \wedge dx^k + \sum_{1 \leq i < j < k \leq n} A_{j,k,i} dx^i \wedge dx^j \wedge dx^k \\
&= \sum_{1 \leq i < j < k \leq n} (A_{i,j,k} - A_{i,k,j} + A_{j,k,i}) dx^i \wedge dx^j \wedge dx^k.
\end{aligned}$$

所以

$$d\omega = I_1 - I_2 = \sum_{1 \leq i < j < k \leq n} (A_{j,i,k} - A_{k,i,j} + A_{k,j,i} - A_{i,j,k} + A_{i,k,j} - A_{j,k,i}) dx^i \wedge dx^j \wedge dx^k.$$

计算：由条件(2) 有

$$\begin{aligned}
&A_{j,i,k} - A_{k,i,j} + A_{k,j,i} - A_{i,j,k} + A_{i,k,j} - A_{j,k,i} \\
&= \frac{\partial b_i^j}{\partial x^k} - \frac{\partial b_i^k}{\partial x^j} + \frac{\partial b_j^k}{\partial x^i} - \frac{\partial b_j^i}{\partial x^k} + \frac{\partial b_k^i}{\partial x^j} - \frac{\partial b_k^j}{\partial x^i} \\
&= \left(\frac{\partial b_i^j}{\partial x^k} - \frac{\partial b_k^j}{\partial x^i} \right) + \left(-\frac{\partial b_i^k}{\partial x^j} + \frac{\partial b_j^k}{\partial x^i} \right) + \left(-\frac{\partial b_j^i}{\partial x^k} + \frac{\partial b_k^i}{\partial x^j} \right) = 0.
\end{aligned}$$

所以 $d\omega = 0$. 所以 ω 是闭形式.

别种计算: 来看

$$\begin{aligned}
&\sum_{i=1}^n \sum_{j=1}^n (b_i^j - b_j^i) dx^i \wedge dx^j = \sum_{1 \leq i < j \leq n} (b_i^j - b_j^i) dx^i \wedge dx^j + \sum_{1 \leq j < i \leq n} (b_i^j - b_j^i) dx^i \wedge dx^j \\
&= \sum_{1 \leq i < j \leq n} (b_i^j - b_j^i) dx^i \wedge dx^j + \sum_{1 \leq i < j \leq n} (b_j^i - b_i^j) dx^j \wedge dx^i \\
&= \sum_{1 \leq i < j \leq n} (b_i^j - b_j^i - (b_j^i - b_i^j)) dx^i \wedge dx^j = 2 \sum_{1 \leq i < j \leq n} (b_i^j - b_j^i) dx^i \wedge dx^j.
\end{aligned}$$

所以

$$2\omega = \sum_{i=1}^n \sum_{j=1}^n (b_i^j - b_j^i) dx^i \wedge dx^j.$$

计算

$$2d\omega = \sum_{i=1}^n \sum_{j=1}^n (d(b_i^j - b_j^i)) \wedge dx^i \wedge dx^j$$

$$\begin{aligned}
&= \sum_{i=1}^n \sum_{j=1}^n \sum_{k=1}^n \left(\frac{\partial b_i^j}{\partial x^k} - \frac{\partial b_j^i}{\partial x^k} \right) dx^k \wedge dx^i \wedge dx^j \\
&= \sum_{i=1}^n \sum_{j=1}^n \sum_{k=1}^n \frac{\partial b_i^j}{\partial x^k} dx^k \wedge dx^i \wedge dx^j - \sum_{i=1}^n \sum_{j=1}^n \sum_{k=1}^n \frac{\partial b_j^i}{\partial x^k} dx^k \wedge dx^i \wedge dx^j \\
&= \sum_{j=1}^n \sum_{i=1}^n \sum_{k=1}^n \frac{\partial b_j^i}{\partial x^k} dx^k \wedge dx^j \wedge dx^i - \sum_{i=1}^n \sum_{j=1}^n \sum_{k=1}^n \frac{\partial b_j^i}{\partial x^k} dx^k \wedge dx^i \wedge dx^j \\
&= \sum_{k=1}^n \sum_{i=1}^n \sum_{j=1}^n \frac{\partial b_k^i}{\partial x^j} dx^j \wedge dx^k \wedge dx^i - \sum_{i=1}^n \sum_{j=1}^n \sum_{k=1}^n \frac{\partial b_j^i}{\partial x^k} dx^k \wedge dx^i \wedge dx^j \\
&= \sum_{k=1}^n \sum_{i=1}^n \sum_{j=1}^n \frac{\partial b_k^i}{\partial x^j} dx^k \wedge dx^i \wedge dx^j - \sum_{i=1}^n \sum_{j=1}^n \sum_{k=1}^n \frac{\partial b_j^i}{\partial x^k} dx^k \wedge dx^i \wedge dx^j \\
&= \sum_{i=1}^n \sum_{j=1}^n \sum_{k=1}^n \frac{\partial b_k^i}{\partial x^j} dx^k \wedge dx^i \wedge dx^j - \sum_{i=1}^n \sum_{j=1}^n \sum_{k=1}^n \frac{\partial b_j^i}{\partial x^k} dx^k \wedge dx^i \wedge dx^j \\
&= \sum_{i=1}^n \sum_{j=1}^n \sum_{k=1}^n \left(\frac{\partial b_k^i}{\partial x^j} - \frac{\partial b_j^i}{\partial x^k} \right) dx^k \wedge dx^i \wedge dx^j = 0
\end{aligned}$$

这里用到微分形式的加法交换律和条件(2). 所以 ω 是闭形式.

[证法3.] 设 $F^i \in C^2(\mathbb{R}^n)$, $i = 1, 2, \dots, n$. 则有

$$F^i(x) = F^i(0) + \int_0^1 \frac{d}{dt} \left(F^i(tx) \right) dt = F^i(0) + \int_0^1 \sum_{k=1}^n \left(\frac{\partial F^i}{\partial x^k} \right) (tx) x^k dt.$$

对比之后, 我们猜可以取

$$F^i(x) = C^i + \int_0^1 \sum_{k=1}^n b_k^i(tx) x^k dt, \quad i = 1, 2, \dots, n. \quad (3)$$

这里 C^i 是常数. 下证由(3)定义的函数 F^i 满足

$$\frac{\partial F^i(x)}{\partial x^j} = b_j^i(x) \quad \forall x \in \mathbb{R}^n, \quad \forall i, j \in \{1, 2, \dots, n\}. \quad (4)$$

事实上由积分号下求导定理和条件(2)有

$$\begin{aligned}
\frac{\partial F^i(x)}{\partial x^j} &= \int_0^1 \sum_{k=1}^n \frac{\partial}{\partial x^j} (b_k^i(tx) x^k) dt = \int_0^1 \sum_{k=1}^n \left\{ \left(\frac{\partial}{\partial x^j} (b_k^i(tx)) \right) x^k + b_k^i(tx) \delta_{j,k} \right\} dt \\
&= \int_0^1 \sum_{k=1}^n \left(\frac{\partial}{\partial x^j} b_k^i \right) (tx) tx^k dt + \int_0^1 \sum_{k=1}^n b_k^i(tx) \delta_{j,k} dt \\
&= \int_0^1 \sum_{k=1}^n \left(\frac{\partial}{\partial x^k} b_j^i \right) (tx) tx^k dt + \int_0^1 b_j^i(tx) dt \\
&= \int_0^1 \frac{d}{dt} (b_j^i(tx)) t dt + \int_0^1 b_j^i(tx) dt \\
&= (b_j^i(tx)) t \Big|_{t=0}^{t=1} - \int_0^1 b_j^i(tx) dt + \int_0^1 b_j^i(tx) dt = b_j^i(tx) t \Big|_{t=0}^{t=1} = b_j^i(x).
\end{aligned}$$

所以(4)成立. 同时由等式(4)和 $b_j^i \in C^1(\mathbb{R}^n)$ 还知由(3)定义的函数 F^i 属于 $C^2(\mathbb{R}^n)$, $i = 1, 2, \dots, n$. 于是再由条件(1)和[证法1]中的运算便知对于 C^2 类的1次形式 $\eta(x) = \sum_{i=1}^n F^i(x)dx^i$ 有 $d\eta = \omega$. 所以 η 是 ω 的一个原函数.

[证法4(不成功).] 考虑

$$F^i(x) = \int_0^{x^1} b_1^i(t, \tilde{x}) dt, \quad x = (x^1, \tilde{x}) \in \mathbb{R} \times \mathbb{R}^{n-1} = \mathbb{R}^n, \quad i = 1, 2, \dots, n.$$

则 $F^i \in C^1(\mathbb{R}^n)$ 且由条件(2)和积分号下求导有

$$\begin{aligned} j = 1, \quad \frac{\partial F^i(x)}{\partial x^1} &= b_1^i(x), \\ j \geq 2, \quad \frac{\partial F^i(x)}{\partial x^j} &= \int_0^{x^1} \frac{\partial b_1^i}{\partial x^j}(t, \tilde{x}) dt = \int_0^{x^1} \frac{\partial b_j^i}{\partial x^1}(t, \tilde{x}) dt = b_j^i(x) - b_j^i(0, \tilde{x}). \end{aligned}$$

因边值 $b_j^i(0, \tilde{x})$ 未必恒为零, 故此法不成功. □